

Applied Mechanistic Interpretability Methods

Reading is not steering: what a probe cannot tell you
about a steering vector

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Abstract



1 Introduction

2 Background: four ways to look inside

3 Exploration: one station per method

What and why. Before choosing a question I ran every method once and small — one notebook per station: patching, steering, probing, sparse autoencoders, plus one pass over a second model. The point was not to find anything. These are single prompts and sample sizes around twenty, orientation rather than statistics. The point was to learn what each instrument measures and where it misleads. Table 1 is the whole tour; the column that matters is the last one. Three of those lessons became the design of Section 5, and the final paragraph says why the experiment then ran on the smaller of the two models.

Station	What I asked	What came out	What it taught
Orientation 01_explore	Does the plumbing behave as documented?	Hook names verified against the library version; BOS carries a residual norm near 3100 against 250 elsewhere; the logit lens is unreadable before layer 6.	Drop position 0 from every mean. Lens probabilities show a direction, not a confidence.
Patching 02_patching	Which heads carry the induction behaviour?	Ablating the five canonical heads destroys 25% of the gap; patching them back restores 60%. No single head restores more than 0.13, and the one with the highest attention score restores 0.05.	Ablation measures necessity, patching sufficiency, and they disagree. A rank by structure is not a rank by effect.
Steering 03_steering	What does a difference-of-means direction do, and at what price?	An S-curve that saturates before the loss does: the metric plateau sits where the loss is already rising. A random vector of the same size leaves the metric flat and costs most of the same loss.	State the dose relative to the layer's residual norm. A random control checks the size of an intervention, not the specificity of its direction.
Probing 04_probing	Is the probe direction different from the difference of means?	On 20 minimal pairs, no: $\cos(\hat{w}, d) \geq 0.95$ at every regularisation strength. The control task reaches 19.4 of 40 held out at training accuracy 1.00.	Minimal pairs cannot separate reading from steering — they leave the probe nothing else to pick up. That test needs heterogeneous data.
SAE 06_sae	Is a sparse feature a cleaner lever than a difference of means?	At equal dose it looks weaker; read at equal effect , both interpolated to +2.00 logits, it is a tie of 0.069 nats against 0.077. The opposite feature costs five times as much.	Compare at equal effect, not at equal dose. An SAE feature is not a better lever, only a differently derived one.
Second model 05_gemma	Does this depend on the quirks of one small model?	Gemma-2-2B matches its reference implementation to $\max \Delta = 0.375$ at a logit scale of 28.6; its BOS sink is in every layer, and the cache costs about 4 MB per token.	The BOS sink and the layer-relative dose are not GPT-2 artifacts. A second model costs memory, not correctness.

Table 1: The exploration tour, one row per notebook in notebooks/. The numbers are orientation at $n \approx 20$, not statistics; what carried into Section 5 is the last column.

What carried over. Three results of the tour became design decisions rather than findings. The probing station failed in the informative direction: the probe and the difference of means were the same direction at every setting, because sentence pairs that differ in one word leave a probe nothing else to separate on — which is why the families in Section 5 vary the template instead of a single word. The SAE station produced the method of comparison: reading two directions at equal effect instead of at equal dose turned a clear-looking loss into a tie, and that reading is what all four sources later get.



The steering station produced the dose rule and the reason it is not optional, since a random vector of the same norm buys most of the same damage while moving nothing. The patching station produced no ingredient, only the habit of not trusting a ranking that was never measured causally.

Which model. The tour covered Gemma-2-2B, and the experiment still runs on GPT-2 small. That was decided by a measurement rather than by preference. The anchor’s own scaling table reports 5 of 23 concepts steerable at 160M parameters, which is a floor effect large enough to make the whole comparison vacuous, so before committing I measured A_{lin} on three candidate families: 0.77, 0.91 and 0.29, each peaking at layer 10 and each exactly zero up to layer 7. Two families well above the anchor’s go/no-go threshold of 0.1 and one in between is enough spread to work with. GPT-2 small also has published sparse autoencoders for all twelve of its layers [1], and the sweep in Section 5 uses one of them per layer. The honest version: this is a budget decision, taken after a pre-measurement instead of before one.

4 Choosing the question

What and why. Between the tour and the experiment sits a decision: out of four methods and a large literature, one question had to be small enough to finish and specific enough to come out wrong. This section says how I surveyed the field, which question I picked and why, which paper I use as an anchor, and which two gaps in it became the measurement in Section 5.

The map. Before choosing I surveyed the four methods of the tour and wrote one row per method: what the field measures, what counts as settled, and which gap is reachable on a laptop. The protocol was fixed before the first search — arXiv and the main proceedings, emphasis on work since 2024, a paper is only included if it reports its own measurement on open weights of at most nine billion parameters, and every entry carries an evidence level recording whether I read the full text, the abstract, or a secondary description. The last rule exists because the search was assisted by a language model, and fabricated references are the systematic failure mode of that: each arXiv identifier was resolved against the arXiv API and checked against title and first author. About 160 titles were screened, 55 arXiv entries kept. Two things stood out. All four methods have the same core question under different names — the corrupt baseline in patching, the KL filter and the integer dose grid in steering, the missing control task in probing [2], [3], the sparsity level in autoencoders [4] — which is to say that the instrument co-determines the finding. And the gaps reachable without training anything are control and baseline work, not new methods.

The cut. Of the candidates that came out of the map I chose the one that asks whether a direction which **reads** a concept well also **steers** it well, measured across concepts, layers and sources of the direction. Three reasons, in descending weight. It sits on the core question above, because reading and steering are two instruments pointed at the same concept and the interesting case is the one where they disagree — a dissociation reported elsewhere as well, for instance a detection direction that stands at 83° to the refusal direction and still steers after rotation [5]. It needs no tooling I did not already have, since probes, difference-of-means directions, SAE latents and the steering loop all exist from the tour. And it has an anchor: a recent paper with a central claim cheap enough to reproduce before extending it. The candidate I dropped last was a dense dose-response study with capability benchmarks, not because the question is uninteresting but because most of its work would be harness building. The control-task candidate did not have to be dropped at all — it fits inside this one as a step.

The anchor. Billa [6] proposes A_{lin} and reports that it predicts where steering will work: across 24 controlled binary families it correlates with steering effect at $\rho = +0.86$ to $+0.91$ and with the choice of layer at $+0.63$ to $+0.92$, while the customary “use a middle layer” heuristic lands on $A_{\text{lin}} = 0$ on



Gemma-2-2B and does nothing there. The paper is candid about its limits: much of that correlation comes from families sitting at the floor, and restricted to $A_{\text{lin}} > 0.1$ it drops to about +0.5 on the smaller models; the tasks are single-token throughout; and steering means a difference of means and nothing else. Two things it does **not** settle are what I could reach. It never states the dose at which it steers, and it reports $\rho(\|d\|, \text{KL}) = +0.96$, so its side-effect axis is very nearly just the norm of the vector it added. And its expectation that SAE features should help where a concept is encoded non-linearly is marked explicitly as untested.

The second work. Tiwari et al. [7] separate decodability from causality: on Gemma-2-9B-it the features geometrically closest to a probe overlap only 11 to 14 per cent with the features that gradients mark as behaviourally relevant. They ablate rather than steer, have no difference-of-means source, and compare at equal set size rather than at equal effect, so what follows is not a rebuild of their study. What I take from them is one check: in their Qwen3-8B experiment 74 per cent of the probe's margin lived in the reconstruction error of the SAE, and where that happens an SAE intervention mostly measures itself. I report that share per family rather than assume it away.

The contribution in one sentence: measure the anchor's diagnosis at equal effect and equal side effect instead of at whatever dose the norm of a difference of means happens to produce, and test its own untested prediction about SAE features while doing so.

Both gaps are cheap. They cost a stated dose axis, a random control and two further sources of the direction — not a new method.

5 Reading versus steering

What and why. This is the experiment the report is built around and the only part with results of its own. It starts from a claim that is easy to state and expensive to check: you can tell in advance, without training anything, whether a steering vector will work in a given layer, by applying the model's own unembedding to that layer and counting how often it already names the answer [6]. Section 5.1 gives the setup and Section 5.2 reproduces the claim on a model an order of magnitude smaller than the ones it was shown on. Section 5.3 is the part I added: the anchor never says how **much** it adds, so I compare four sources of a steering direction at equal side effect instead of at whatever dose each one happens to have. Section 5.4 says what survives that comparison, Section 5.5 what this setup cannot decide. If you read one table, read Table 4.

5.1 Setup

The families follow the anchor's design: two answer classes, a shared sentence template, and an single-token answer — which matters, because it makes the effect of an intervention a change in one probability, with no decoding choices entering the measurement. I built six families and picked them by measuring A_{lin} beforehand rather than by guessing (Table 2). Four carry signal on GPT-2 small; two do not, and they stay in the set for exactly that reason: if a method reports an effect on parity, the method is wrong, not the model. The intervention always pushes the second class towards the first class's answer, so the direction keeps one sign throughout.

Where possible a family uses several sentence templates, because with a single one a probe can separate the classes on the subject word alone. continent and parity have one each, and their probe numbers should be read with that in mind.

Family	Answer A / B	Prompts	Templates	Peak A_{lin}
pronoun	she / he	36 + 36	3	0.99
temperature	cold / hot	36 + 36	3	0.68
daynight	day / night	16 + 16	16	0.31
size	small / big	36 + 36	3	0.25
continent	Europe / Asia	12 + 12	1	0.04
parity	odd / even	18 + 18	1	0.00

Table 2: The six families, ordered by peak A_{lin} . Answers are single tokens and carry the leading space GPT-2 tokenises them with. The last two are floor cases: the model cannot answer them at all, so nothing should be steerable there.

Three ways of reading a concept off a layer are measured per layer:

- A_{lin} — the model’s own unembedding applied to the residual stream after each layer, argmax compared with the answer token, share correct. This is the logit lens [8] used as a **predictor** rather than as a description, and it needs no training and one forward pass.
- A **logistic probe** on the same residual stream, fitted per layer, scored on a held-out quarter, with a control task on shuffled labels beside it [2], [9].
- **AUROC of the difference-of-means direction** — how well the single direction between the class means separates held-out examples.

And one way of writing one in: add a direction at every prompt position of the second class’s prompts. Its **effect** is ΔP , the change in probability of the first class’s answer token; its **side effect** is the Kullback–Leibler divergence of the next-token distribution on 50 unrelated prompts. Both are the anchor’s measures, so its numbers and mine are comparable. KL on unrelated prompts is a narrow notion of damage — it says nothing about generation quality — but it is cheap, it means the same thing for every family, and it needs no judge.

Everything runs on GPT-2 small [10] (124M parameters, 12 layers, $d_{\text{model}} = 768$) through Transformer-Lens [11], seed 0, on a laptop GPU. The two notebooks run top to bottom in about 20 seconds and about three and a half minutes, write their numbers to `results/`, and reproduced those files byte for byte when re-run for this report.

5.2 The anchor, reproduced

The anchor reports its correlations on five models from Pythia-2.8B to Llama-8B. GPT-2 small is smaller than all of them, so reproducing the **pattern** here tests the claim rather than confirming it politely: a measure like this could easily be an artifact of scale. It is not. For every family except pronoun, A_{lin} is exactly zero in layers 0 to 7 and rises only in the last quarter — the same shape the anchor reports in layers 18 to 24 of a 2B model. pronoun is the exception and jumps to 0.64 at layer 6. Within families, $\rho(A_{\text{lin}}, \Delta P)$ across the twelve layers is +0.48 to +0.90, inside the anchor’s band of +0.63 to +0.92.

The probe fails as a layer selector, and it fails worse than “worse than A_{lin} ”. For continent and pronoun the held-out probe reads 1.00 in **every one of the twelve layers**: it has no variance left to rank layers with, so its correlation with steering effect is not weak, it is undefined. Where it is defined it changes sign by family, +0.74 on daynight against −0.39 on size. A saturated probe is not a poor predictor of steerability; it is not a predictor.

continent is the sharpest case, and the one the subtitle refers to:

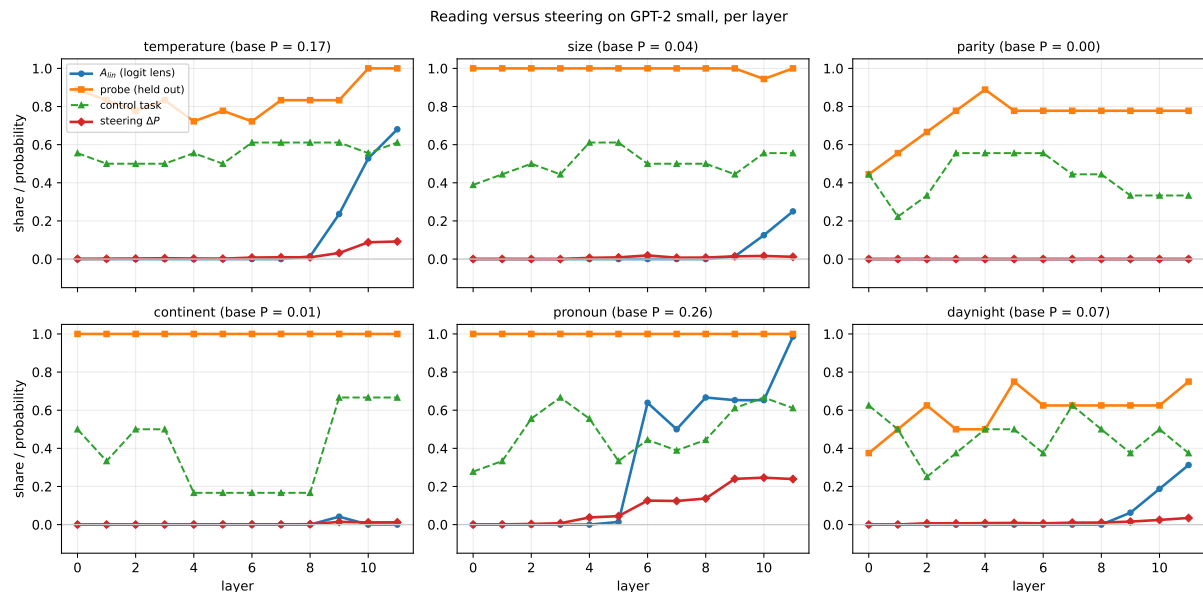


Figure 1: Reading against steering per layer, one panel per family. A_{lin} (circles) rises late; the probe (squares) is flat and high wherever the family has one template; the control task (triangles) shows the probe is not merely memorising; steering ΔP (diamonds) follows A_{lin} , not the probe. Source: `notebooks/07_reading_vs_steering.ipynb`.

On continent the probe reads the concept perfectly in all twelve layers, against a control task at 0.17 to 0.67 — so this is not the probe memorising labels. A_{lin} peaks at 0.042, and steering moves the answer by at most +0.014. The concept is present, linearly readable, and unsteerable.

Two further claims of the anchor hold. The common “use a middle layer” heuristic loses exactly where A_{lin} says it should: temperature gains +0.092 at layer 11 against +0.008 at layer 6, and daynight +0.035 against +0.007. And the go/no-go threshold survives — parity and continent stay below 0.05 peak A_{lin} and produce $\Delta P \leq +0.014$, which is what such a threshold is for.

Then the crack. The anchor adds the raw difference of means at coefficient 1, and across these six families and twelve layers that vector’s norm ranges from 0.5 to 58.4, while the mean residual norm of GPT-2 small grows from 76 in layer 0 to 644 in layer 10. Coefficient 1 is therefore not one dose but 72 of them, and an efficiency measure of $\Delta P / \text{KL}$ compares concepts at whatever dose their difference of means happens to have.

5.3 Four sources of a direction, at equal side effect

What and why. Same six families, same seed, same 50 unrelated prompts; three things change. The dose is stated and held fixed relative to the layer. Four sources of the direction are compared instead of one, and one of them is random. And the comparison is read at equal side effect, because a direction that buys its effect with more damage is not a better direction. All of it serves one number: how much of the anchor’s reported correlation is its dose tracking its own predictor.

Dose. All four directions are scaled to unit norm and added as α times the mean residual norm of that layer. This is the choice the whole comparison rests on, so it is worth one sentence of justification: the residual stream of GPT-2 small grows by a factor of 8.5 across depth (76 $\rightarrow$ 644, then down to 489 in layer 11), so one absolute dose is a heavy intervention early and a light one late. A relative dose makes “the same amount” mean the same thing in every layer.



Sources. The difference of means; the probe's weight vector; one SAE decoder row; and a random direction as the floor. The random one is the best of five draws rather than a single draw — one random vector is a sample, and a floor built from a sample moves. The SAEs are the public GPT-2 residual-stream set [1] (24 576 features per layer, $32\times$ expansion). I checked the hook mapping instead of trusting the folder names: `blocks.{l+1}.hook_resid_pre` is bit-identical to `blocks.{l}.hook_resid_post`, so folder $l + 1$ holds the SAE for layer l , and SAEs exist for layers 0 to 10. **Layer 11 has none, and layer 11 is where A_{lin} peaks for all four carrying families.** The SAE arm is blind at the most interesting layer — a property of the available weights, not of the method.

The feature is selected on the training half with a criterion fixed before looking: fire on at least 60% of one class and at most 25% of the other, with the sign fixed to the class the intervention pushes towards. How often that fails is worth reporting, because the alternative is to keep searching until something is found. In layer 6 nothing passes for four of six families and the fallback is the largest mean gap, with held-out AUROC of 0.38 to 0.79 — two of them below chance. In layer 10 the criterion is met everywhere except parity, with held-out AUROC of 0.88 to 1.00 for five families and daynight the exception at 0.40.

Readouts. A twelve-layer scan at fixed $\alpha = 0.1$, then a ten-point dose sweep (α from 0.01 to 0.4) in layer 6 — the middle-layer heuristic — and layer 10, the deepest layer with an SAE. Each sweep yields two numbers, both fixed in advance: ΔP interpolated at $\text{KL}^* = 0.1$ nats, and the KL needed to reach $\Delta P^* = +0.05$. The first is interpolated in log KL because KL is monotone in dose. The second takes the first crossing on the rising branch, because ΔP is **not** monotone: past some dose the effect collapses as the model stops producing a sensible distribution at all.

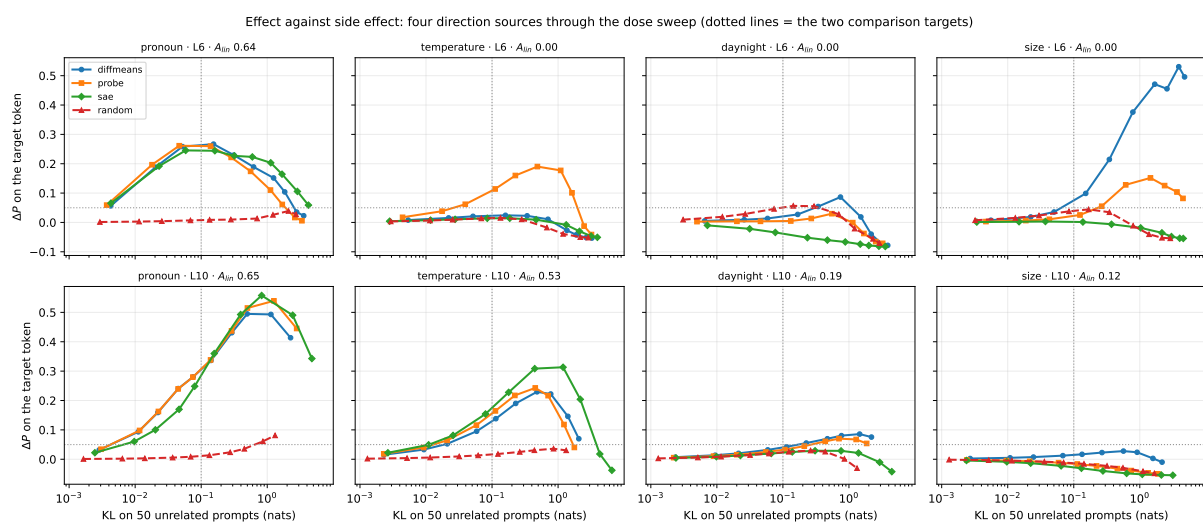


Figure 2: Effect against side effect for the four sources, top row layer 6, bottom row layer 10, log KL on the x-axis. The dotted lines are the two comparison targets. Each curve is one direction through ten doses; the random floor (dashed) is the best of three draws per point. Source: `notebooks/08_direction_sources.ipynb`.

Family	Layer	diff-of-means	probe	SAE	random
pronoun	6	+0.263	+0.260	+0.245	+0.008
pronoun	10	+0.306	+0.307	+0.285	+0.011
temperature	6	+0.023	+0.109	+0.014	+0.015
temperature	10	+0.130	+0.156	+0.175	+0.017
size	6	+0.075	+0.022	+0.002	+0.039
size	10	+0.015	-0.015	-0.028	-0.013
daynight	6	+0.021	+0.005	-0.038	+0.051
daynight	10	+0.040	+0.032	+0.023	+0.024

Table 3: Effect ΔP at the same side effect (KL = 0.1 nats), best per row in bold. The two floor families are omitted: parity is exactly 0.000 in every cell, continent at most +0.023.

5.4 What survives the fairer comparison

Table 4 is the answer to the question this experiment was built for. The anchor’s **implicit** dose — the norm of its difference of means divided by the residual norm of the layer — ranges from 0.003 to 0.119 across families and depth, and within each carrying family it correlates with A_{lin} at +0.76 to +0.91. Its dose rises exactly where its predictor rises. Holding the dose fixed, the within-family correlation between A_{lin} and steering effect drops from +0.80, +0.57, +0.48, +0.90 to +0.56, -0.10, +0.54, +0.24.

Family	Peak A_{lin}	Implicit dose	$\rho(A_{\text{lin}}, \text{dose})$	ρ at coeff. 1	ρ at $\alpha = 0.1$
temperature	0.68	0.004 – 0.085	+0.84	+0.80	+0.56
size	0.25	0.006 – 0.117	+0.76	+0.57	-0.10
pronoun	0.99	0.005 – 0.086	+0.91	+0.90	+0.54
daynight	0.31	0.005 – 0.067	+0.76	+0.76	+0.24
continent	0.04	0.005 – 0.119	+0.31	+0.48	-0.22
parity	0.00	0.003 – 0.013	flat	flat	flat

Table 4: The dose confound, in numbers. Column 3 is the dose the anchor applies without stating it; column 4 shows it tracking the anchor’s own predictor. Columns 5 and 6 are the same correlation at the anchor’s dose and at a fixed relative dose. parity has $A_{\text{lin}} = 0$ in every layer, so there is no rank order to correlate.

That looks like a refutation, and my first reading of it was exactly that, because at a fixed dose the ranking of the families does not merely weaken — it inverts, with size moving from last place to first. I held that reading back, for one reason: at a fixed dose the four families do not pay the same side effect, so the fixed-dose ranking is no fairer than the anchor’s. Read at an equal KL budget instead, the order comes out as pronoun > temperature > size > daynight against the predicted pronoun > temperature > daynight > size — one adjacent swap.

The apparent collapse of the ranking is itself a dose artifact. The weakening of the correlation is not. The anchor’s predictor is real and its reported strength is inflated by a dose it never states.

Four further findings, all from Figure 2 and Table 3.

The unstated dose is not even the best one. With the same direction in layer 10, pronoun reaches $\Delta P = +0.495$ at $\alpha = 0.2$, against +0.246 at the anchor’s coefficient 1. The dose is not only unreported, it is left on the table.



Where the direction comes from matters less than which layer and how much. In layer 10 the three principled sources land within 0.03 of each other on pronoun and within 0.05 on temperature. The large differences in the grid are between layers, not between derivations — awkward for a literature that mostly compares methods of extraction.

The anchor’s untested SAE prediction does not hold uniformly. Its third regime — high A_{lin} , so the concept is already output-aligned and an SAE feature should add little — holds for pronoun (+0.285 against +0.306) and fails for temperature in layer 10, where the SAE direction wins outright (+0.175 against +0.130).

Explained variance is blind to the concept. The SAE reconstruction explains 92 to 99% of the variance of the residual stream, while 35 to 99% of the class separation along the reading direction sits in the reconstruction **error** — the share Tiwari et al. measure at 74% on another model, another SAE and other concepts [7]. Replacing one layer by its own reconstruction costs the model its answer: accuracy on temperature falls from 0.68 to 0.35, on size from 0.25 to 0.07, on daynight from 0.31 to 0.09 (pronoun holds, 0.99 to 0.86). A number that high and a loss that large in the same object is this report’s clearest argument for reading a quality metric as a claim about what it measures, not about what one hopes it measures.

Finally, the controls earn their place. The random floor is not zero and it wins once: on daynight in layer 6 it beats all three principled sources at equal side effect (+0.051 against +0.021, +0.005 and −0.038). And parity stays at exactly 0.000 across four sources, twelve layers and ten doses — any pipeline reporting an effect there would have been reporting its own noise. [5] find the same dissociation from the other side on a larger model, where a perfectly separating detection direction stands at 83° to the direction that actually steers.

5.5 What this setup cannot decide

The directions are fitted on the same prompts that are then steered, as in the anchor; there is no held-out steering set, so the effect sizes are upper bounds. The families have 24 to 72 examples against 768 dimensions, so the held-out quarter is 4 to 18 prompts — which is why no probe number here is quoted to three digits. The cross-family correlation rests on six points and is an illustration, not evidence; the anchor uses 24. I also left out the anchor’s second measure, A_{mlp} , so its middle regime — present but not linearly readable — rests on the probe alone here, and a failing probe never proves absence [3]. The random floor is the best of five draws, not a distribution, and the SAE arm cannot see layer 11, where four of six families peak.

None of this is fatal to the conclusion, which is a comparison between directions measured the same way rather than an absolute claim about steerability. It does mean that every number in this section is a statement about GPT-2 small with these six families; the anchor’s own result across five larger models is the reason to think the pattern is more than that.

6 What I take from this

7 Working method and use of AI

References

- [1] J. Bloom, “Open Source Sparse Autoencoders for All Residual Stream Layers of GPT2-Small.” 2024.

- [2] J. Hewitt and P. Liang, “Designing and Interpreting Probes with Control Tasks,” in *Proceedings of EMNLP-IJCNLP*, 2019.
- [3] Y. Belinkov, “Probing Classifiers: Promises, Shortcomings, and Advances,” *Computational Linguistics*, vol. 48, no. 1, pp. 207–219, 2022.
- [4] H. Cunningham, A. Ewart, L. Riggs, R. Huben, and L. Sharkey, “Sparse Autoencoders Find Highly Interpretable Features in Language Models,” *arXiv preprint arXiv:2309.08600*, 2023.
- [5] Galeone and others, “Perfect Detection, Failed Control,” *arXiv preprint arXiv:2606.24952*, 2026.
- [6] J. Billa, “Predicting Where Steering Vectors Succeed,” *arXiv preprint arXiv:2604.15557*, 2026.
- [7] D. Tiwari, C. Davis, S. Sinha, T. Weaver, A. Shah, and M. Chaudhary, “Decodability is Not Causality: Dissociating Probe Readouts from Behavioral Drivers via SAE Decomposition,” *arXiv preprint arXiv:2609.18080*, 2026.
- [8] nostalgebraist, “Interpreting GPT: The Logit Lens.” 2020.
- [9] G. Alain and Y. Bengio, “Understanding Intermediate Layers Using Linear Classifier Probes,” *arXiv preprint arXiv:1610.01644*, 2017.
- [10] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever, “Language Models Are Unsupervised Multitask Learners,” technical report, 2019.
- [11] N. Nanda and J. Bloom, “TransformerLens.” 2022.

A Appendix